Artefact

AI Powered Resume Generator

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Artefact

Entries produced during the software development lifecycle can be referred to as artefacts. Artefacts are essentially any documentation or physical representation of a piece of software. The different types of artefacts include items such as diagrams, models, documents, source code, wireframes and test plans which provide support to developers and stakeholders for the conduct of their respective roles throughout the course of the project. Artefacts define requirements, design structure and how the software will function as all these inputs provide a foundation that creates a framework for the developer and its stakeholders to work within therefore providing for a more structured and clear manner of how to proceed with the creation of the software. If sufficient artefacts are unavailable to the software development team, there is a potential that the process will not be conducted in an orderly fashion and as such is viewed as building a structure without plans therefore the possibility for error and miscommunication would increase significantly. (Pressman, 2020)

In addition, the use of artefacts in the agile methodology is essential in enabling the iterative and incremental approach to software development. Agile provides for continuous collaboration between developers, testers and clients in the creation of a product and in doing so provides artefacts such as product backlogs, sprint backlogs and wireframes to maintain transparency and a shared understanding. Agile differs from the waterfall methodology in that agile allows for simultaneous development and testing enabling for improved and faster feedback loops and reduced risk of missed requirements and ultimately an increase in the quality of the final product and the satisfaction of the users with the end result. (Sommerville, 2016)

Wireframes are a frequently used early stage artifact to demonstrate potential user interfaces and layout of a system. The creation of wireframes, before any coding begins, allows for the designers to concentrate on usability and user experience prior to creating production ready version of a product, enabling an opportunity for designers to enhance the design in the earliest phases of development at a cost effective manner. In addition, there are test plans that we will also create and use as part of the quality assurance process for validation of the reliability of the system. The combination of both black box and white box testing will help to validate that the functional requirements of the system meet the expected user experience as well verify that the internal logic of the system operates correctly without defects. (Garrett, 2011) (Myers, 2011)

Product Backlog

A backlog is a prioritized and updated collection of tasks to manage an organization's software design and findings greater to those of all parties (stakeholders) who have agreed to develop an application according to those parties stated. Responses early in the application development process are initial requests for clarification from the user; therefore, these clarifications may lead to new tasks for resolution. (Schwaber, 2020)

Product BackLog - AI Powered Resume Generator				
Epic	**User Story**	**Acceptance Criteria**	**Priority**	**Sprint**
User	I want to be able to sign up so that I can create an account and use the system	Given that I am on the signup page, when I fill in valid details and submit, then my account should be created	Must be	Sprint 1
User	I want to be able to log in so that I can access my dashboard	Given that I have a registered account, when I enter valid credentials, then I should be logged in and redirected to the dashboard	Must be	Sprint 1
User	I want to be able to log out so that no one else can access my account	Given that I am logged in, when I click the logout button, then I should be logged out successfully	Must be	Sprint 1
User	I want to reset my password so that I can regain access if I forget it	Given that I click on "Forgot Password", when I enter my registered email, then I should be able to reset my password	Should be	Sprint 1
Resume Creation & Management				
Epic	**User Story**	**Acceptance Criteria**	**Priority**	**Sprint**
User	I want to create a resume by entering my personal and career details so that AI can generate my resume	Given that I enter required details, when I submit the form, then an AI-generated resume should be created	Must be	Sprint 2
User	I want to upload my existing resume so that the system can analyze and improve it	Given that I upload a PDF resume, when processing completes, then the resume content should be extracted successfully	Must be	Sprint 2
User	I want to select a resume template so that my resume looks professional	Given that templates are available, when I select one, then the resume layout should update accordingly	Must be	Sprint 2
User	I want to download my resume so that I can use it for job applications	Given that my resume is finalized, when I click download, then it should be downloaded as a PDF or DOCX	Must be	Sprint 2

AI Optimization & Job Matching				
Epic	**User Story**	**Acceptance Criteria**	**Priority**	**Sprint**
User	I want AI suggestions so that my resume becomes ATS compliant	Given that AI analysis is complete, when suggestions are shown, then resume optimization tips should be displayed	Must be	Sprint 3
User	I want to see a resume score so that I can understand my resume quality	Given that the resume is analyzed, when the score is calculated, then it should be displayed clearly	Should be	Sprint 3
User	I want to compare my resume with a job description so that I can see how well it matches	Given that I paste a job description, when analysis is run, then matching skills and missing keywords should be shown	Must be	Sprint 3
User	I want AI to generate a cover letter so that I can apply faster	Given that resume and job description exist, when I click generate cover letter, then a customized cover letter should be created	Should be	Sprint 3
System Administration				
Epic	**User Story**	**Acceptance Criteria**	**Priority**	**Sprint**
Admin	I want to manage user accounts so that the system remains secure and controlled	Given that I am an admin, when I view user accounts, then I should be able to activate or deactivate users	Must be	Sprint 2
Admin	I want to monitor resume generation activity so that system usage can be analyzed	Given that users generate resumes, when viewed in admin panel, then usage statistics should be visible	Should be	Sprint 3

Figure 1 Product Backlog of AI Powered Resume Generator

Sprint Backlog

Sprint Backlog - Sprint 1(AI Powered Resume Generator)			
Epic	**User Story**	**Acceptance Criteria**	**Priority**
Admin	I want to be able to add admin login so that I can manage the system	Given that I have added an admin login page, when I enter valid admin credentials, then I should be logged into the admin dashboard	Must be
Admin	I want to be able to manage user accounts so that I can control system access	Given that I am logged in as admin, when I view users, then I should be able to activate or deactivate user accounts	Must be
Admin	I want to monitor resume generation activity so that I can analyze system usage	Given that users generate resumes, when I view the admin dashboard, then usage activity should be displayed	Should be
User	I want to be able to sign up so that I can create an account and use the system	Given that I am on the signup page, when I fill in valid details and submit the form, then my account should be created successfully	Must be
User	I want to be able to log in using my credentials so that I can access the system	Given that my account exists, when I enter valid email and password, then I should be logged in and redirected to my dashboard	Must be
User	I want to be able to log out so that no one else can use my account	Given that I am logged in, when I click the logout button, then I should be logged out and redirected to the login page	Must be
User	I want to reset my password so that I can regain access if I forget it	Given that I click on the forgot password option, when I enter my registered email, then I should be able to reset my password	Should be
User	I want secure authentication so that my data remains protected	Given that I log in or register, when authentication occurs, then my session should be secured using tokens	Must be

Figure 2 Sprint Backlog of AI Powered Resume Generator

Functional Decomposition Diagram (FDD)

A Functional Decomposition Diagram breaks down the overall system into smaller functional modules and sub functions to clearly illustrate how each component contributes to the complete system functionality. (Jekyll, 2018)

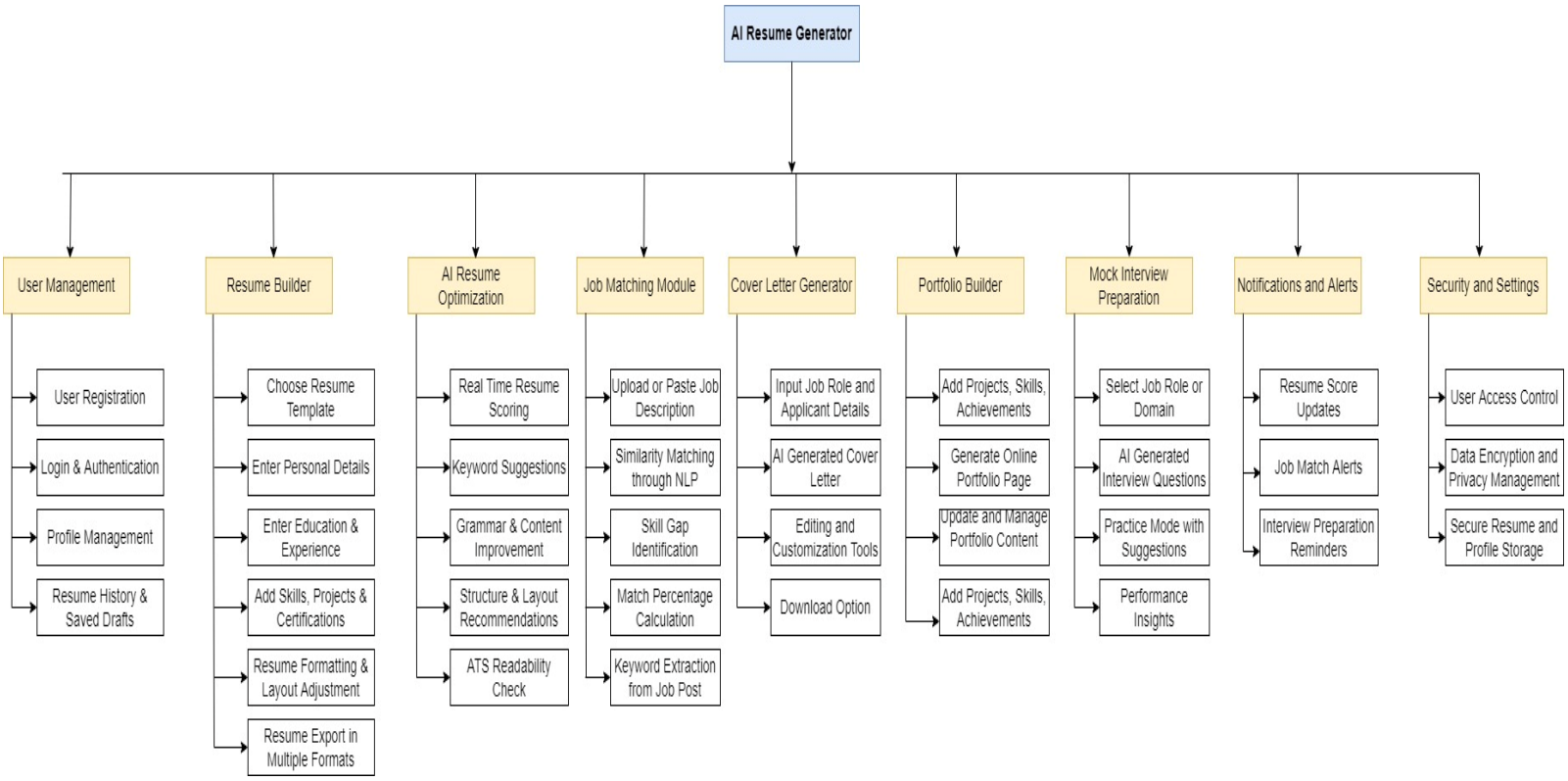

Figure 3 FDD of AI Powered Resume Generator

Legend:

- **UM**: User Management
- **RB**: Resume Builder
- **AIO**: AI Optimization & Analytics
- **JMM**: Job Matching Module (NLP)
- **CLG**: Cover Letter Generator
- **MP**: Mock Interview Preparation

Requirement Types:

- **F:** Functional Requirements
- **NF:** Non-Functional Requirements
- **UR:** Usability Requirements

User Management (UM)

Req. Code	Requirement Description	Use Case	Type	MoSCoW
UM-F-1.0	The system shall allow users to register and login using Email, Google, or LinkedIn OAuth.	User Authentication	F	Must Have
UM-F-1.1	The system shall implement JWT (JSON Web Tokens) to manage secure user sessions and API access.	Secure Login (PB-01)	F	Must Have
UM-NF-1.1	All user passwords must be hashed using Bcrypt before being stored in the database.	Data Security	NF	Must Have

UM-UR-1.1	The login interface must provide a "Password Visibility" toggle for mobile-friendly input.	User Registration	UR	Should Have

Table 1User Management Table for AI Powered Resume Generator

Unified Modeling Language (UML)

Unified Modeling Language (UML) is a standardized visual language used to model and document the structure and behavior of software systems, showing elements like actors, processes, and interactions to improve understanding and communication. (Larman, 2005)

Activity Diagram

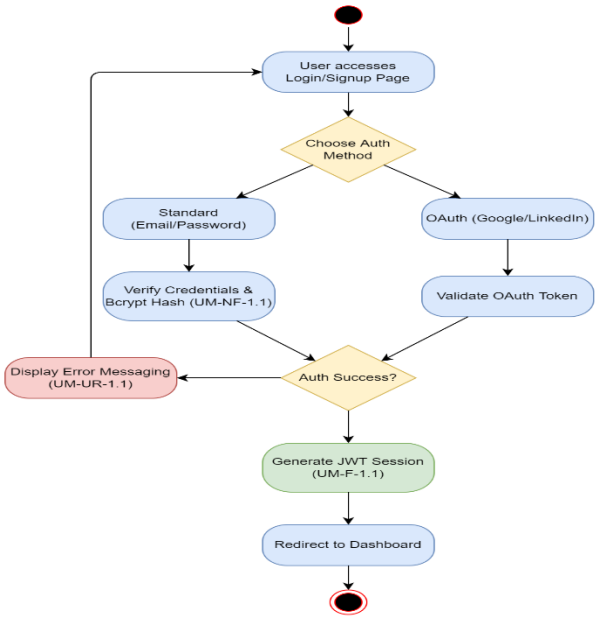

Figure 4 Activity Diagram of UM

Sequence Diagram

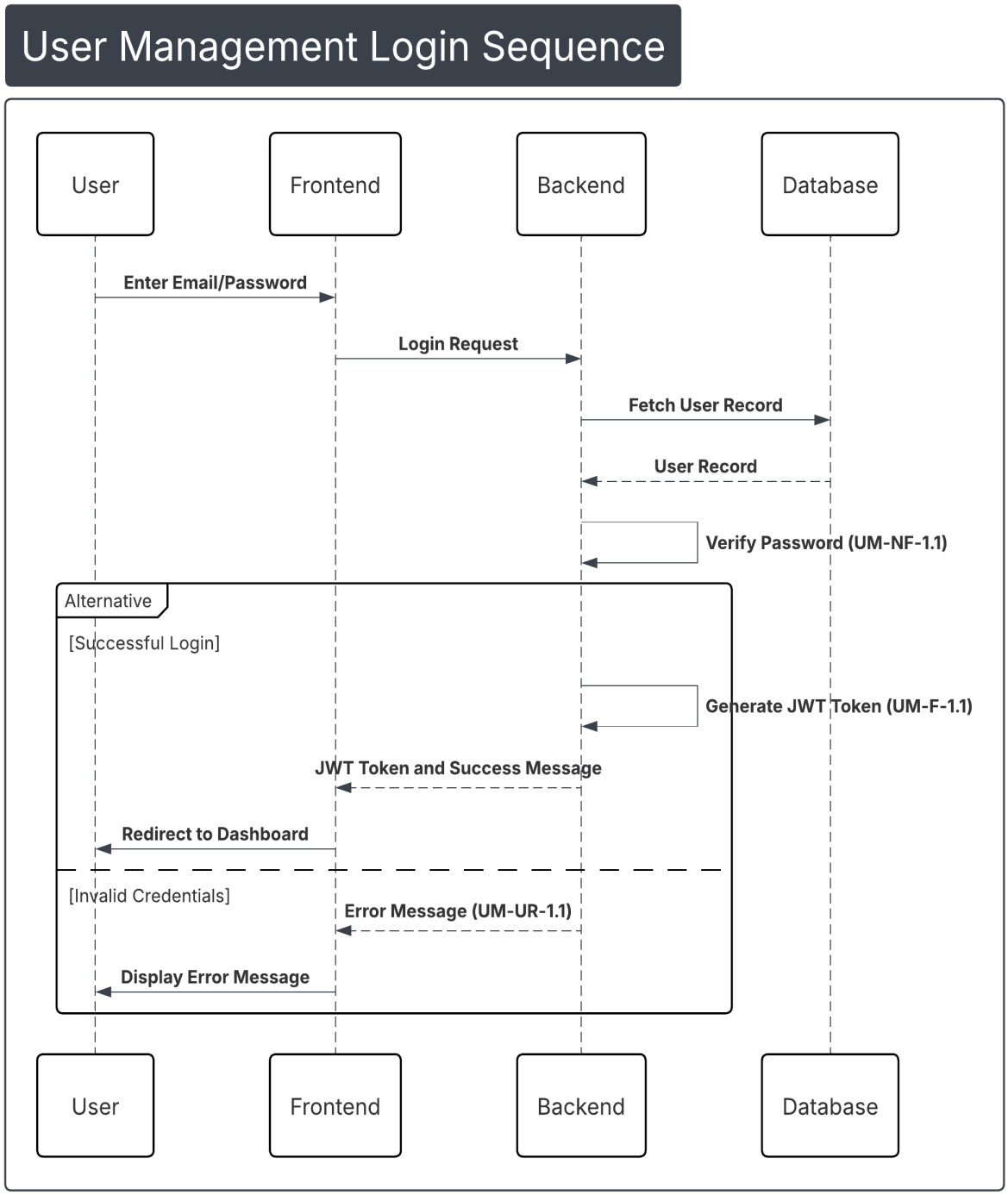

Figure 5 Sequence Diagram of UM

Use Case Diagram

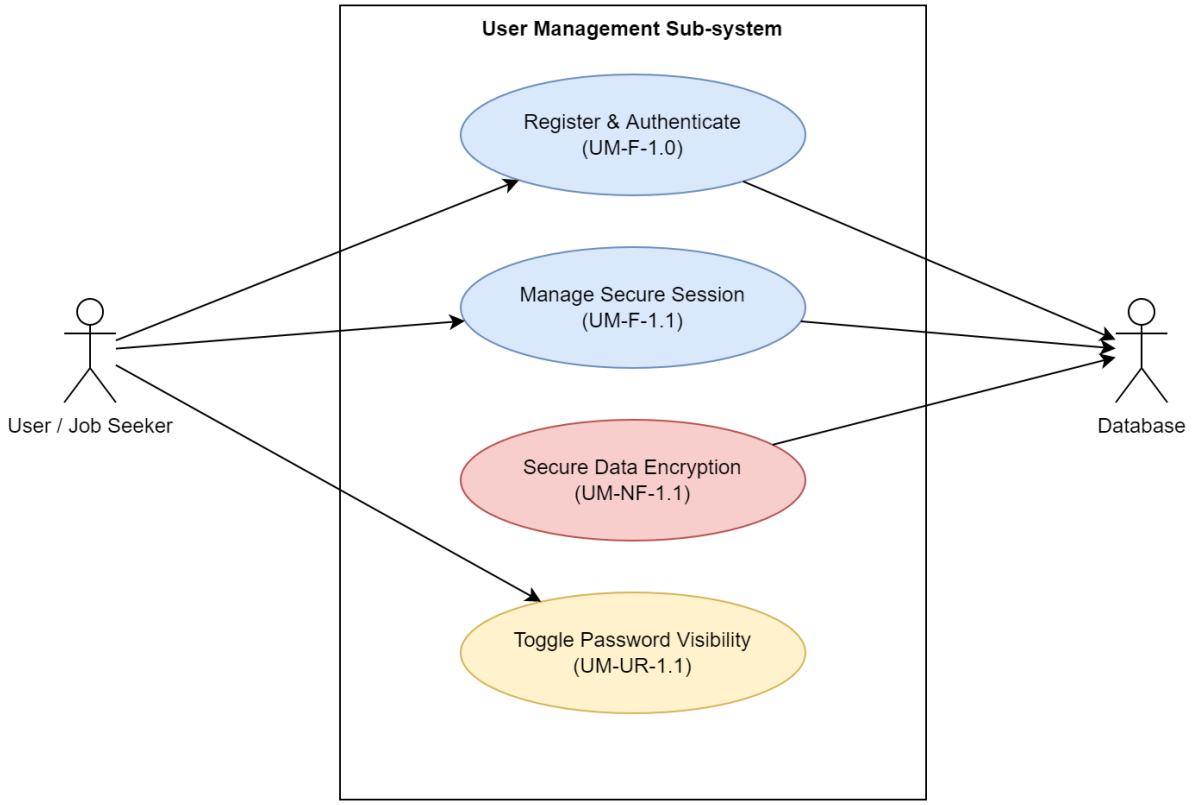

Figure 6 Use Case Diagram of UM

Resume Builder (RB)

Req. Code	Requirement Description	Use Case	Type	MoSCoW
RB-F-2.0	The system shall provide a React-based engine to preview resume changes in real-time	Template Preview (PB-02)	F	Must Have
RB-F-2.1	The system shall allow users to export the final resume in PDF format using server-side rendering.	Resume Export (PB-02)	F	Must Have
RB-NF-2.1	The PDF generator must	Internationalization	NF	Should Have

	support UTF-8 encoding to allow for special characters and symbols.			
RB-UR-2.1	The editor should use a "What You See Is What You Get" (WYSIWYG) layout to simplify data entry.	Data Input	UR	Must Have

Table 2 Table of Resume Builder(RB)

Activity Diagram

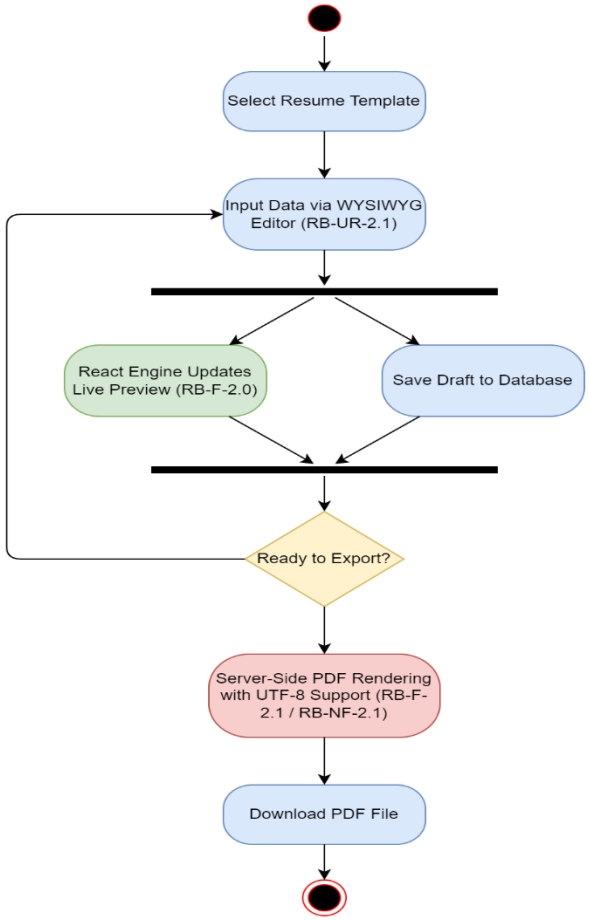

Figure 7Activity Diagram of RB

Sequence Diagram

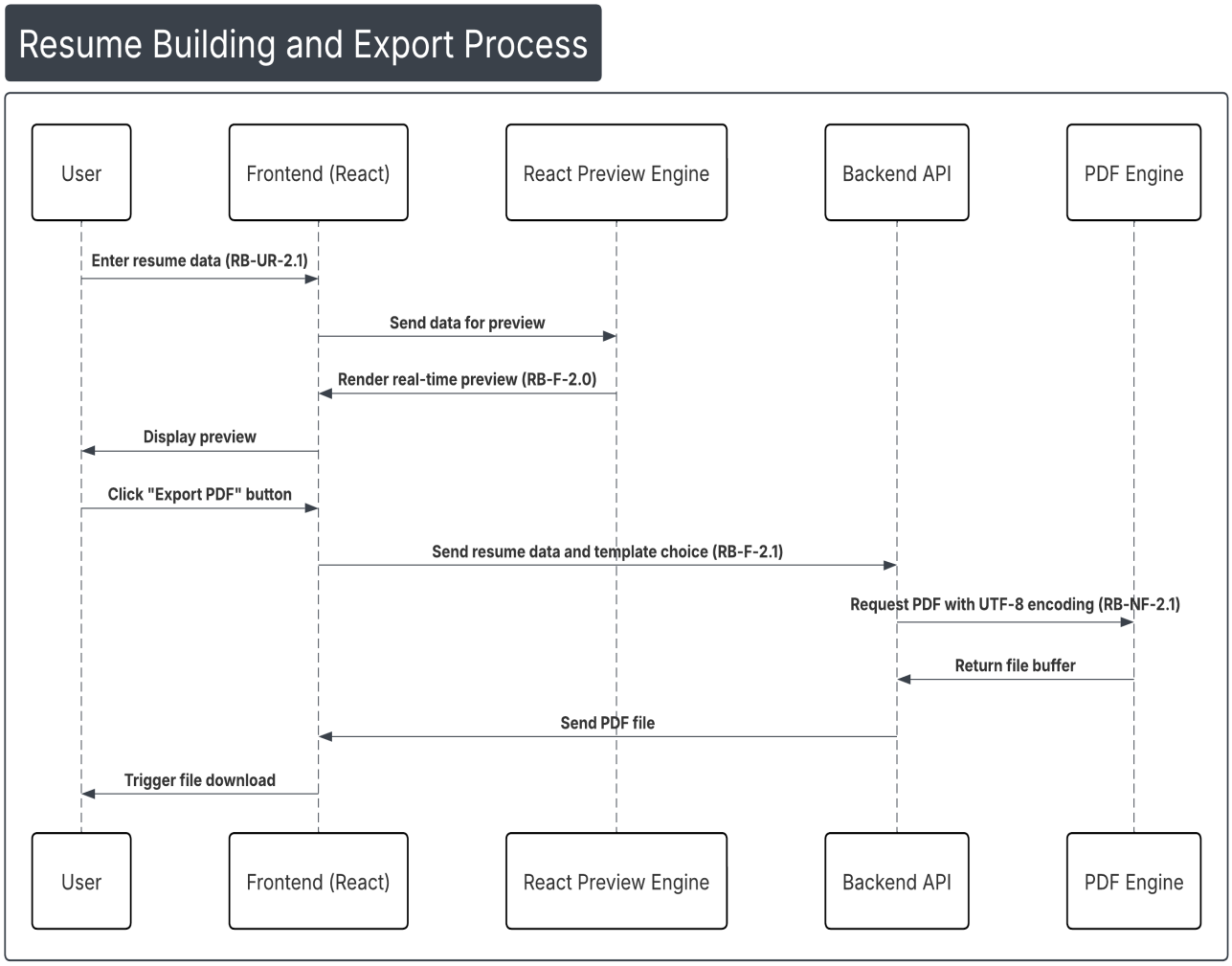

Figure 8 Sequence Diagram of RB

Use Case Diagram

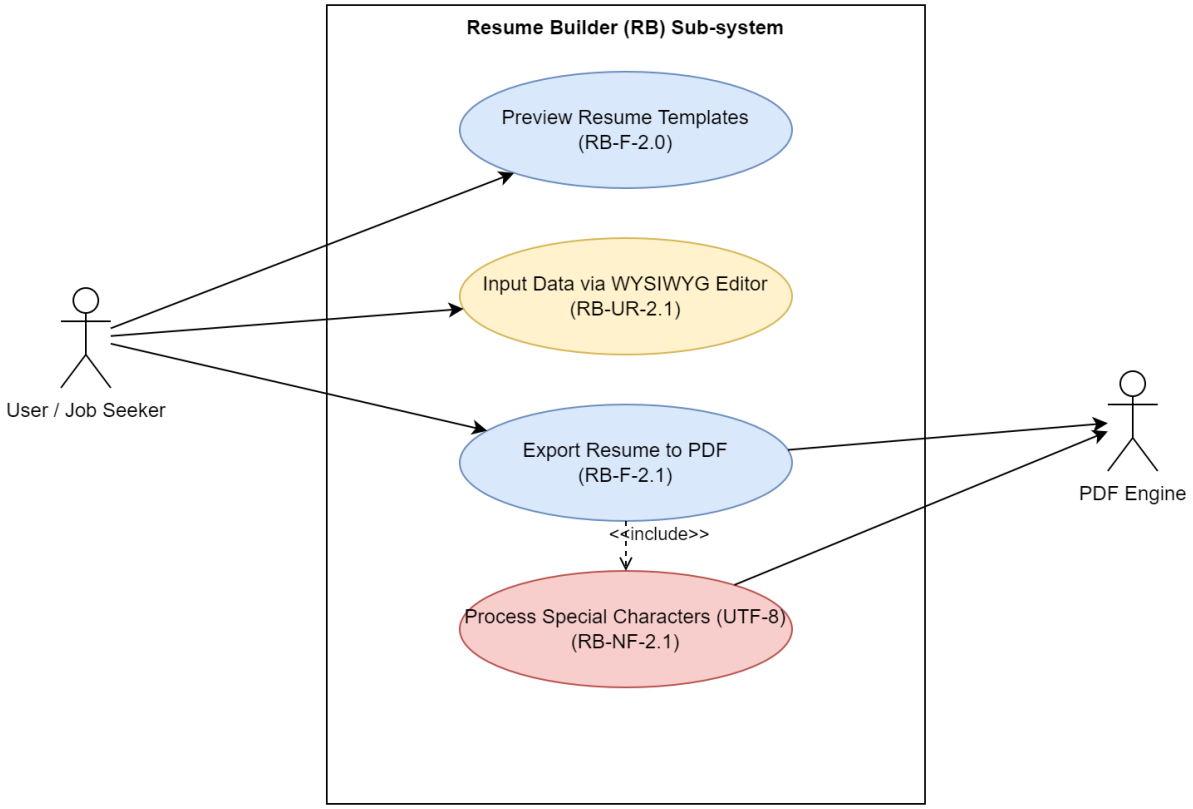

Figure 9 Use Case Diagram of RB

AI Optimization & Job Matching (AIO/JMM)

Req. Code	Requirement Description	Use Case	Type	MoSCoW
AIO-F-3.0	The system shall integrate OpenAI API to analyze and score resume content against job standards.	AI Scoring (PB-03)	F	Must Have
JMM-F-3.1	The system shall extract keywords from a Job Description to perform similarity matching.	Keyword Integration (PB-03)	F	Must Have

JMM-F-3.2	The system shall generate a visual chart or list identifying specific missing skills (Gaps).	Skill Gap UI (PB-04)	F	Should Have
AIO-NF-3.1	The matching engine must complete the analysis of a 2-page resume in under 5 seconds.	Performance	NF	Should Have

Table 3 Table of AIO/JMM

Activity Diagram

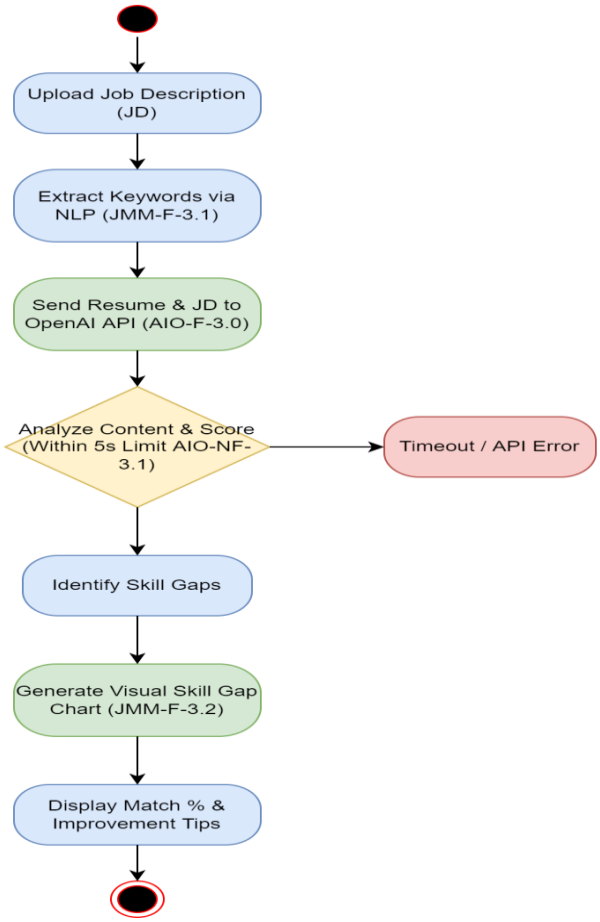

Figure 10 Activity Diagram of AIO/JMM

Sequence Diagram

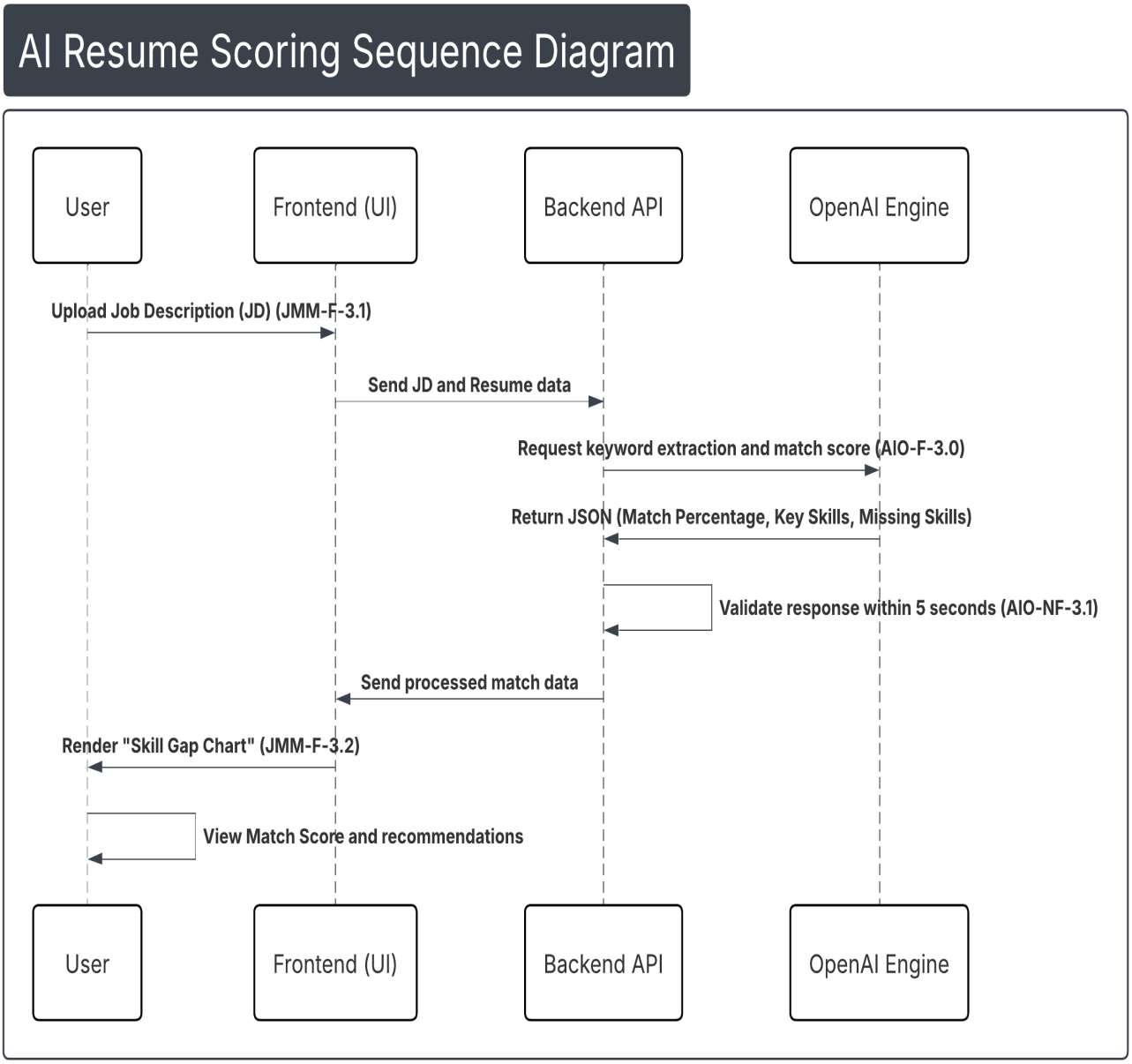

Figure 11 Sequence Diagram of AIO/JMM

Use Case Diagram

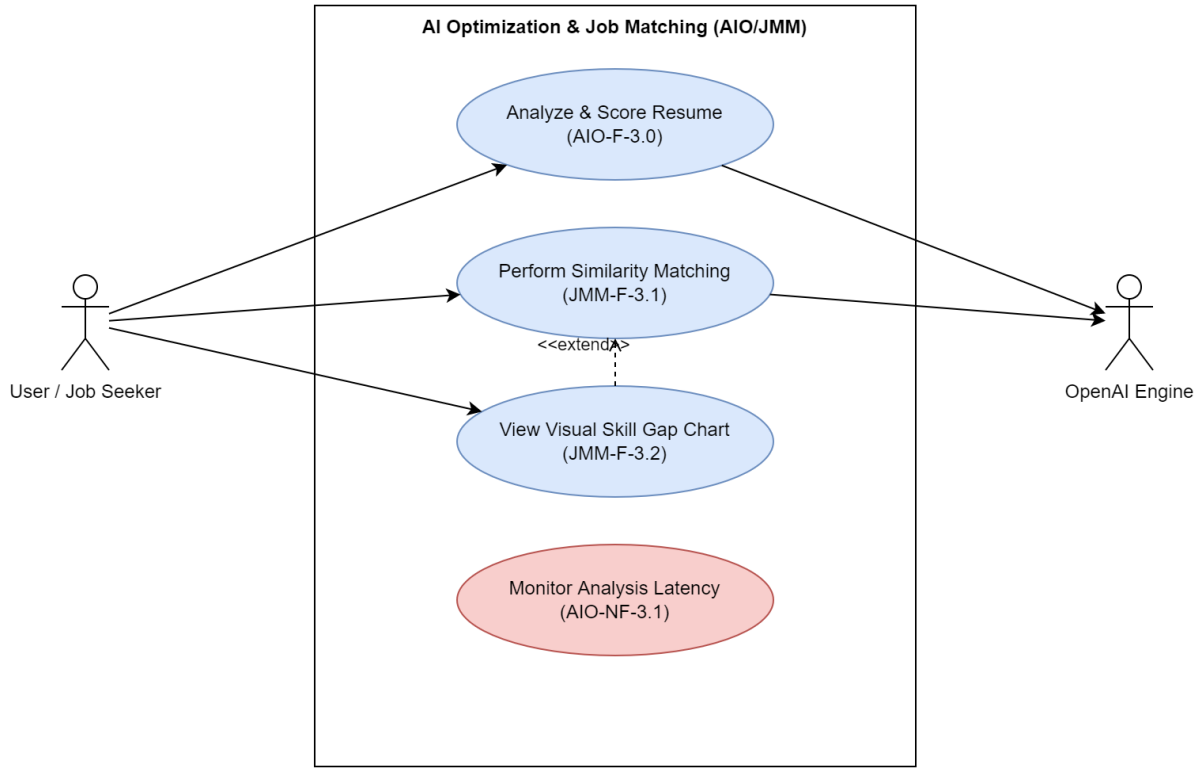

Figure 12 Use Case Diagram of AIO/JMM

Mock Interview Preparation (MP)

Req. Code	Requirement Description	Use Case	Type	MoSCoW
MP-F-4.0	The system shall generate tailored interview questions based on the user's specific experience.	Interview Prep (PB-05)	F	Could Have
MP-F-4.1	The system shall provide a text-based practice mode with AI-generated performance insights.	Practice Mode	F	Could Have
MP-UR-4.1	Interview questions should	Interview Simulation	UR	Could Have

	be presented one by one to simulate a real interview environment.			

Table 4 Table of MP

Activity Diagram

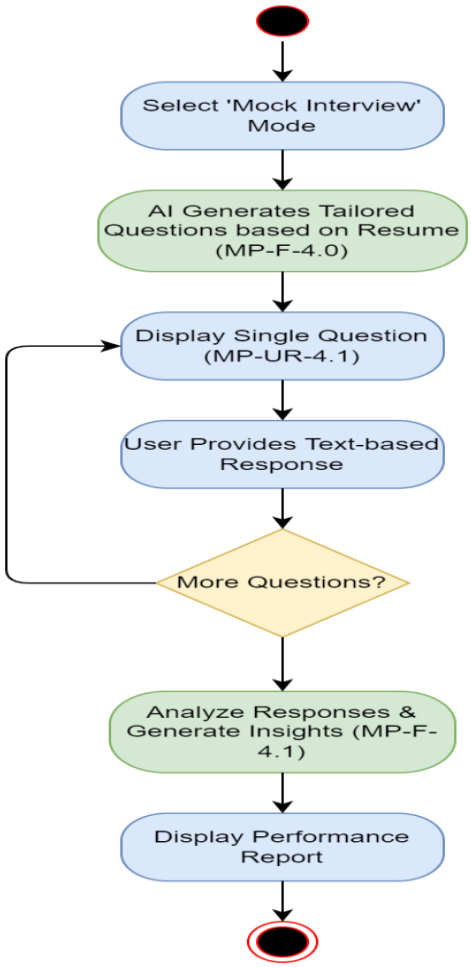

Figure 13 Activity Diagram of MP

Sequence Diagram

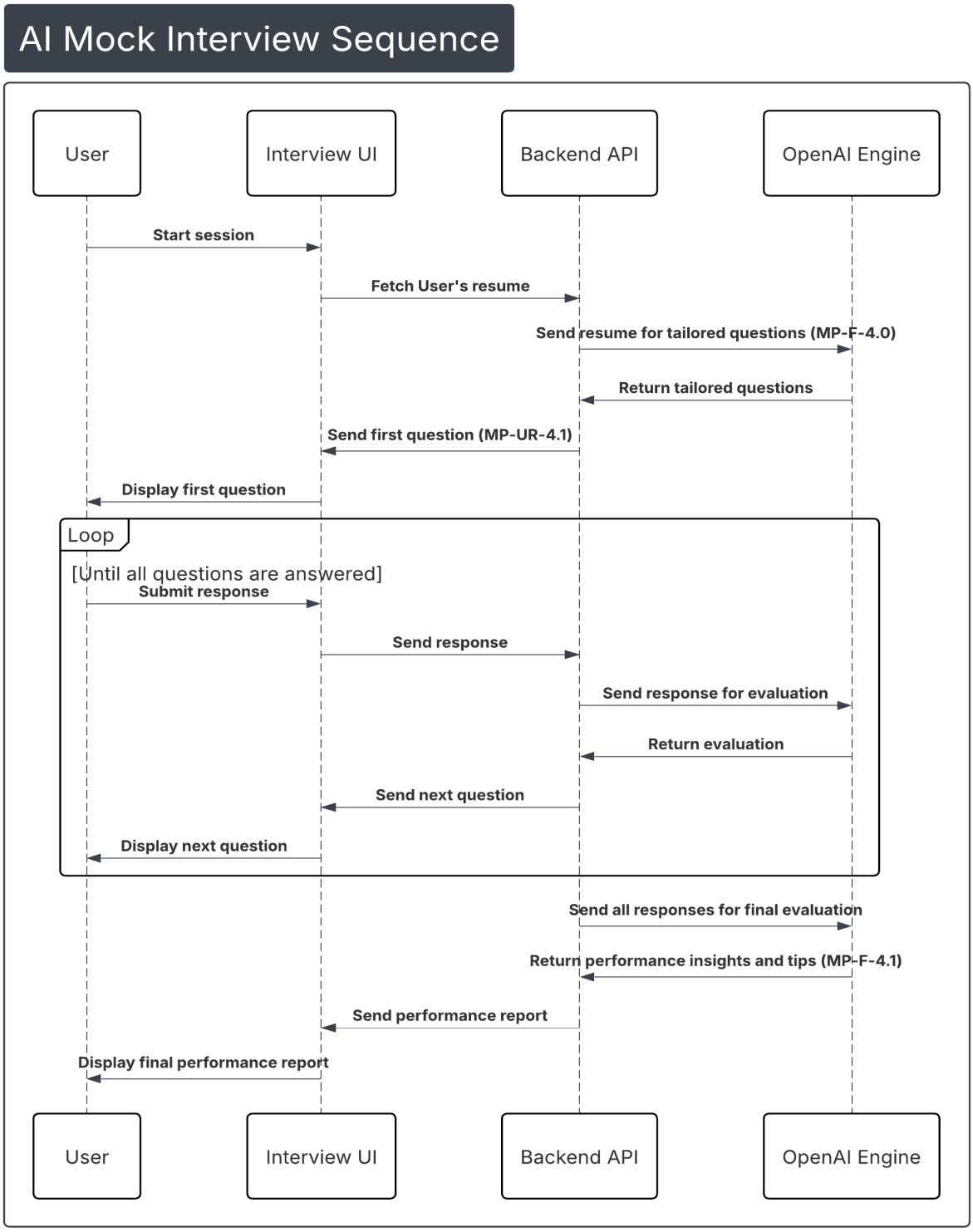

Figure 14 Sequence Diagram of MP

Use Case Diagram

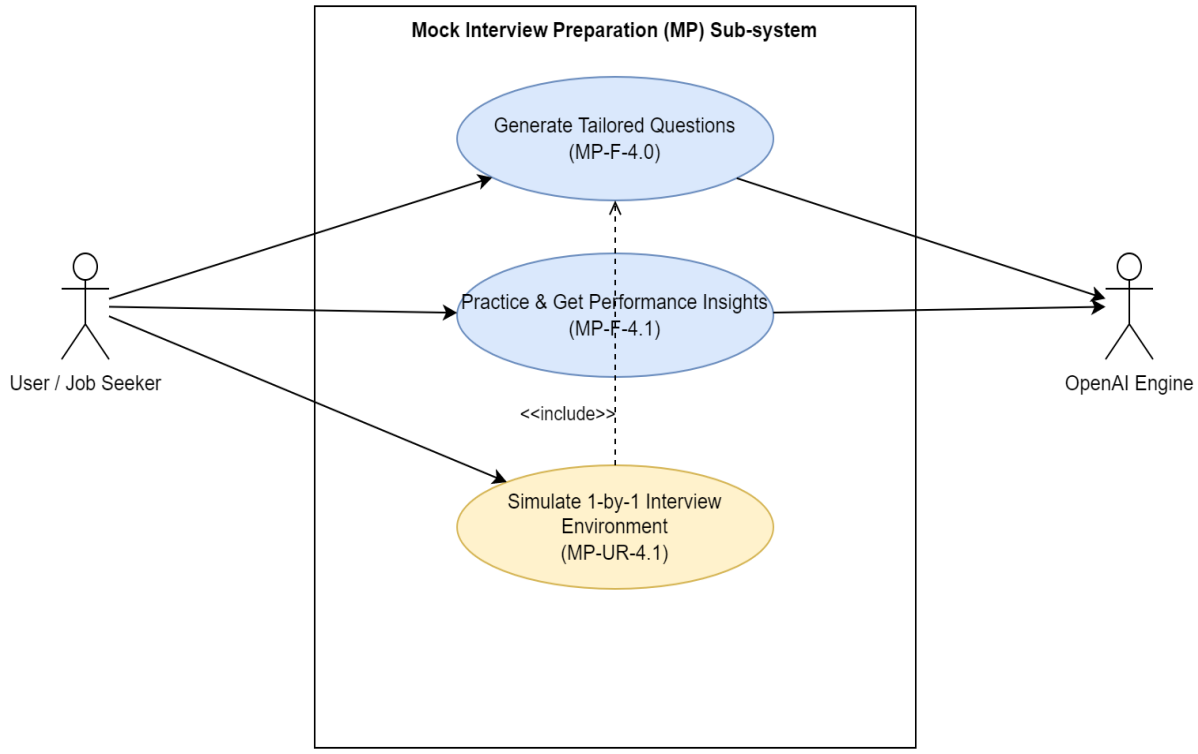

Figure 15 Use Case Diagram of MP

Class Diagram

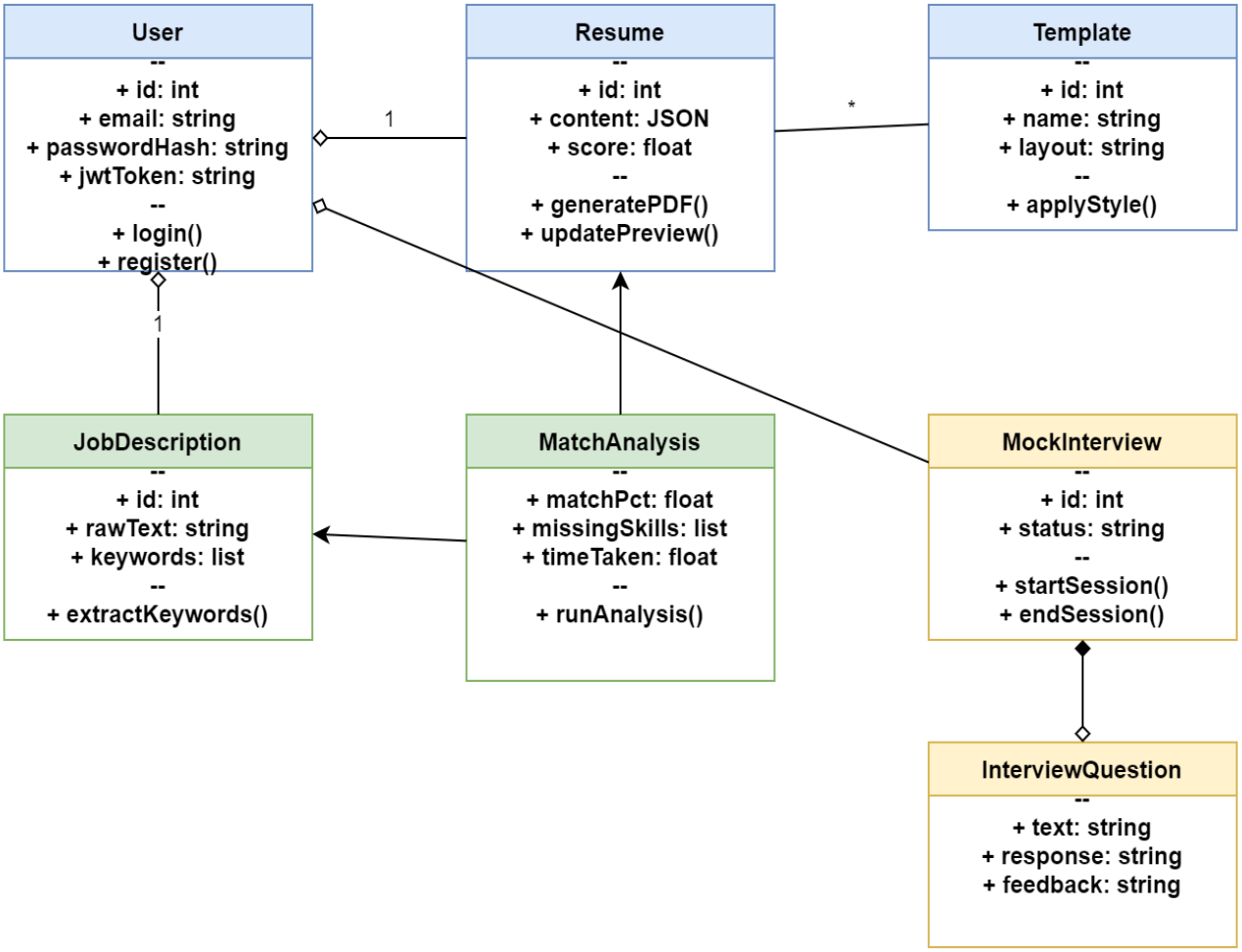

Figure 16 Class Diagram of AI Powered Resume Generator

Entity Relationship Diagram

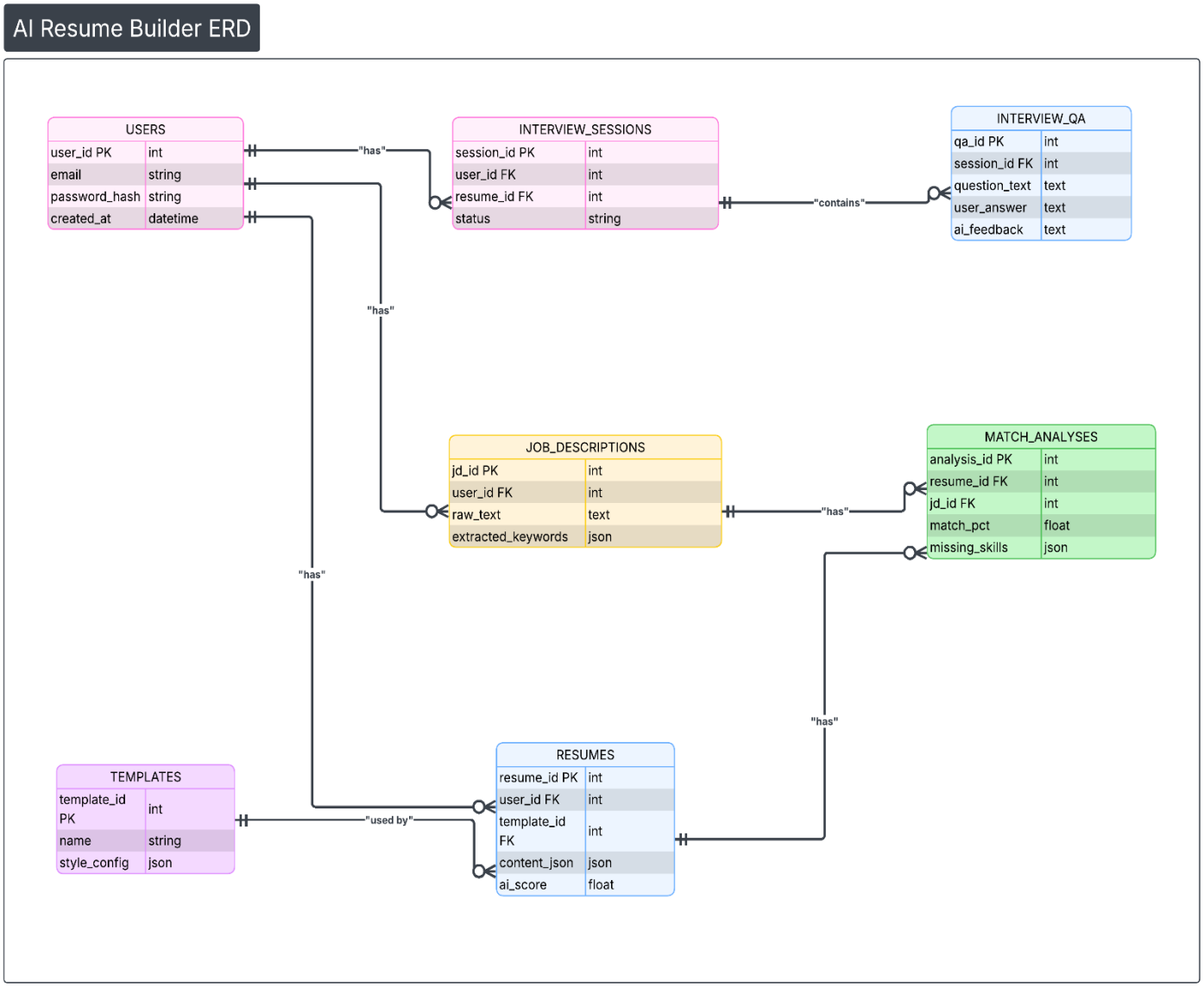

Figure 17 Entity Relationship Diagram of AI Powered Resume Generator

Testing Case

Black Box Testing:

Black box testing is a software testing technique in which the tester evaluates the functionality of the system by providing inputs and examining outputs without any knowledge of the internal code or design**.** (Hamilton, 2021)

White Box Testing:

White box testing is a software testing technique in which the tester evaluates the internal logic, structure, and code implementation of the system while designing and executing tests. (Hamilton, 2021)

TC_ID	Requirement ID	Keywords	Priority	Testcase Title	Description	Precondition	Steps	Test Data	Expected Result	Remark
WBT_001	UM-NF-1.1	Security	Must be	Password Hashing Logic	Verify that the password is encrypted using Bcrypt before DB insertion.	User registration service is active.	1. Intercept the registration request. 2. Inspect the password field in the database.	Pwd: "testPassword123"	Database stores a 60 character Bcrypt hash, not plain text.	Pending
WBT_002	UM-F-1.1	Logic	Must be	JWT Token Signature	Validate that the JWT secret key is used to sign tokens correctly.	User is logged in.	1. Capture the Authorization header. 2. Attempt to decode without the secret key.	Header: Bearer [Token]	Token cannot be manipulated or verified without the backend secret.	Pending
WBT_003	RB-F-2.1	Functionality	Must be	PDF UTF-8 Rendering	Verify the internal function handles special symbols during PDF generation.	PDF library is initialized.	1. Pass a string with "€" and "ñ". 2. Trigger the generatePDF() function.	Input: "José €100"	The generated PDF displays symbols correctly without distortion.	Pending
WBT_004	AIO-F-3.0	API	Must be	OpenAI API JSON Parsing	Ensure the backend correctly parses the JSON response from OpenAI.	API key is valid.	1. Send mock JD/Resume data. 2. Verify the runAnalysis() function extracts specific keys.	Mock API Response	Backend correctly identifies "match_pct" and "missing_skills".	Pending
WBT_005	AIO-NF-3.1	Performance	Should be	Latency Calculation	Test the internal timer for the 5-second analysis limit.	Performance monitoring service active.	1. Trigger AI matching. 2. Measure execution time of the matching module.	Resume (2 pages)	Analysis completes and returns data in < 5000ms.	Pending

Table 5 Test Case for AI Powered Resume Generator

Wireframe

A wireframe is a low-fidelity visual blueprint that outlines the layout, structure, and basic elements of a webpage or application interface without focusing on design details or styling. (UX, 2021)

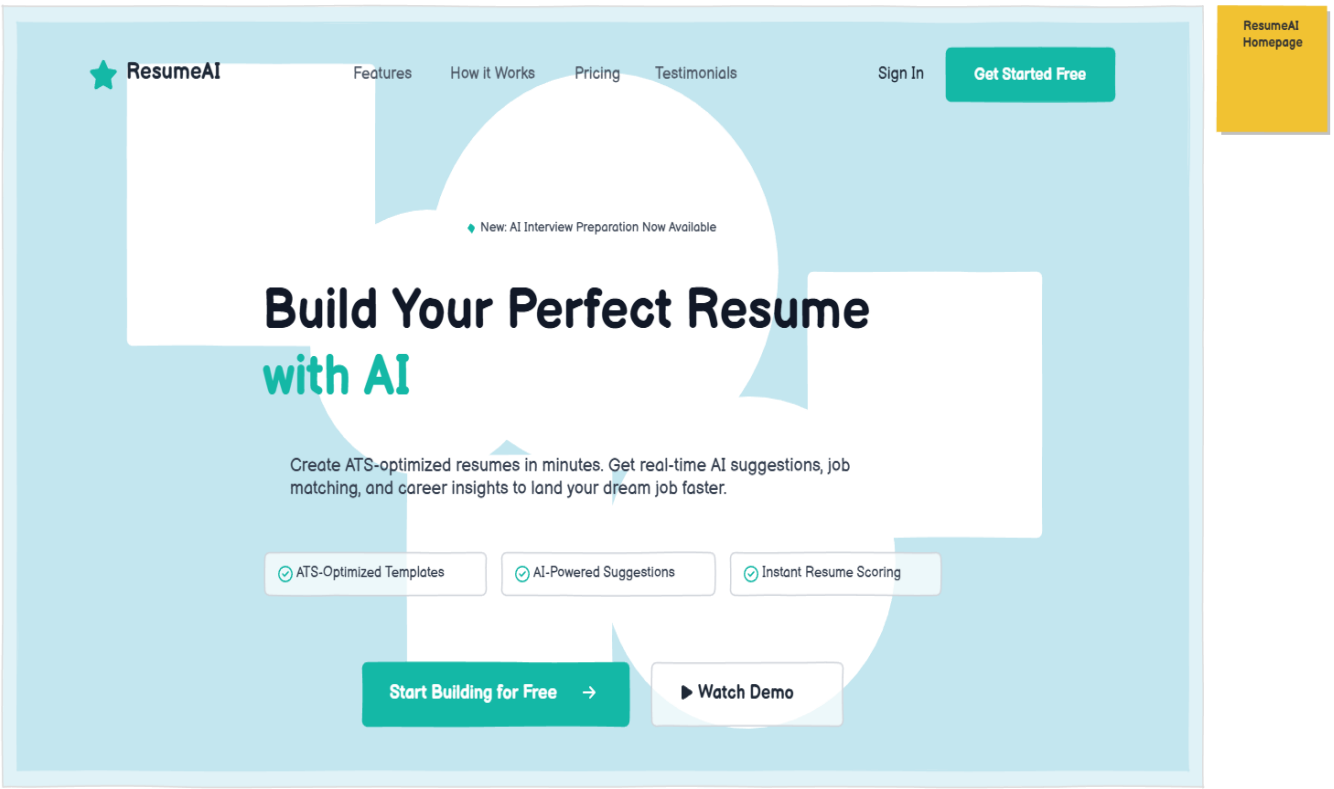

Figure 18 Wireframe of Resume Homepage

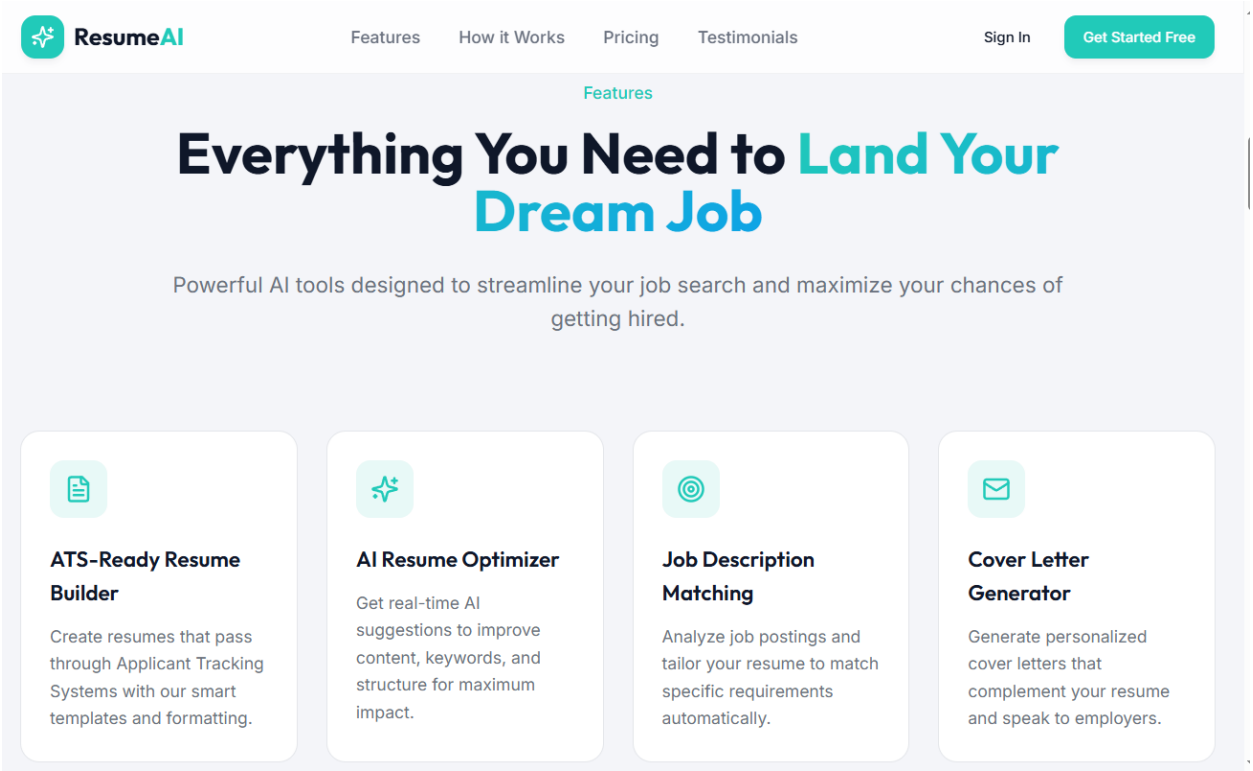

Figure 19 Wireframe of Feature page

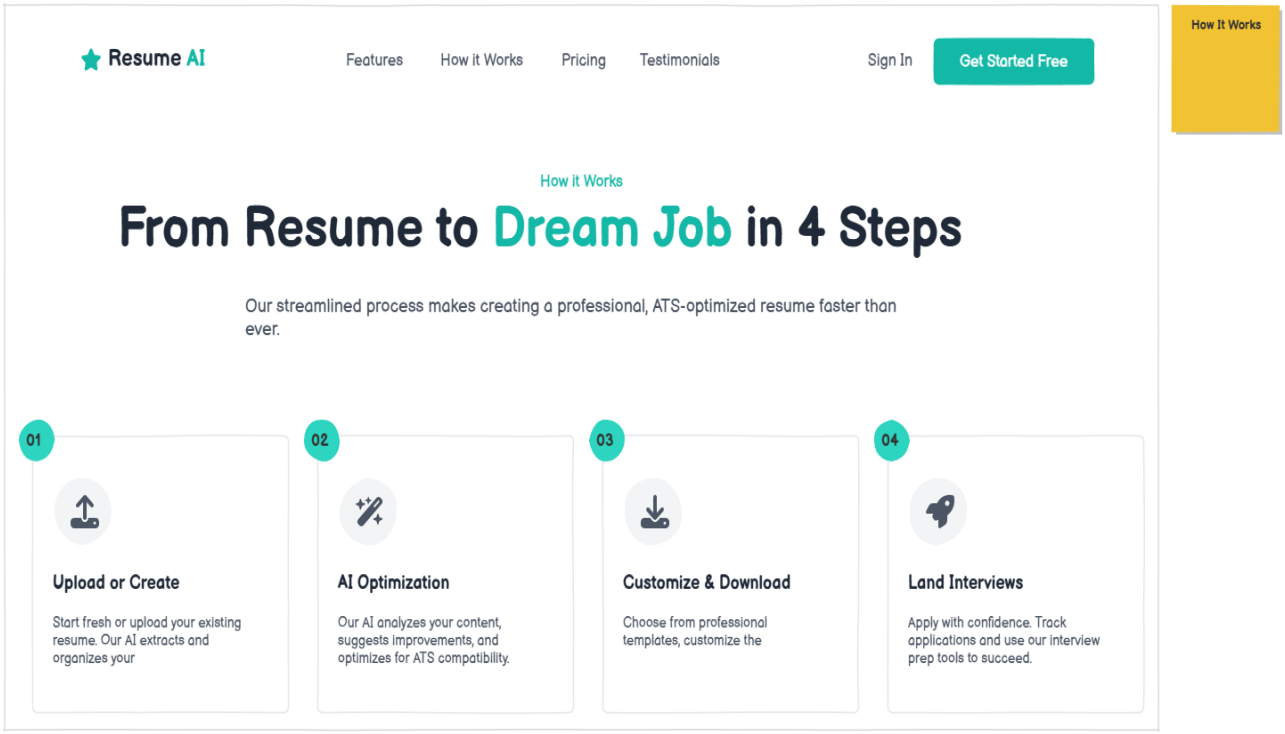

Figure 20 Wireframe of How it Works page

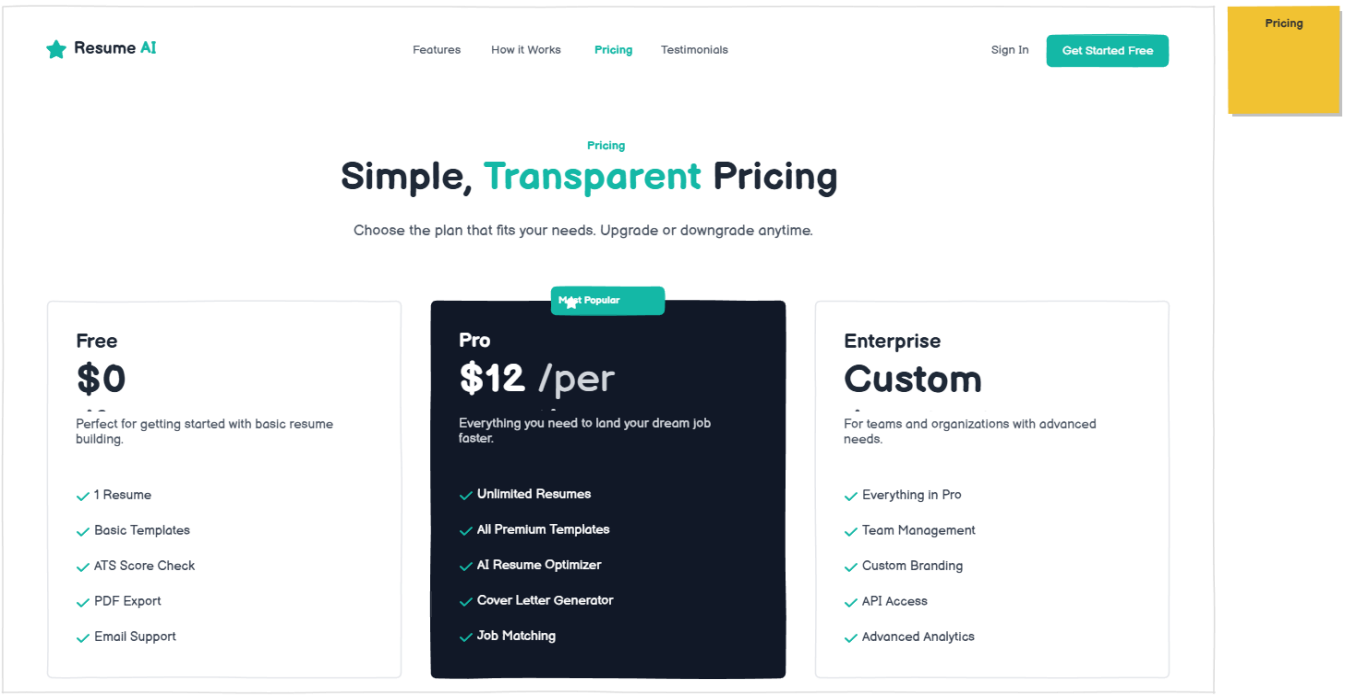

Figure 21 Wireframe of Pricing page

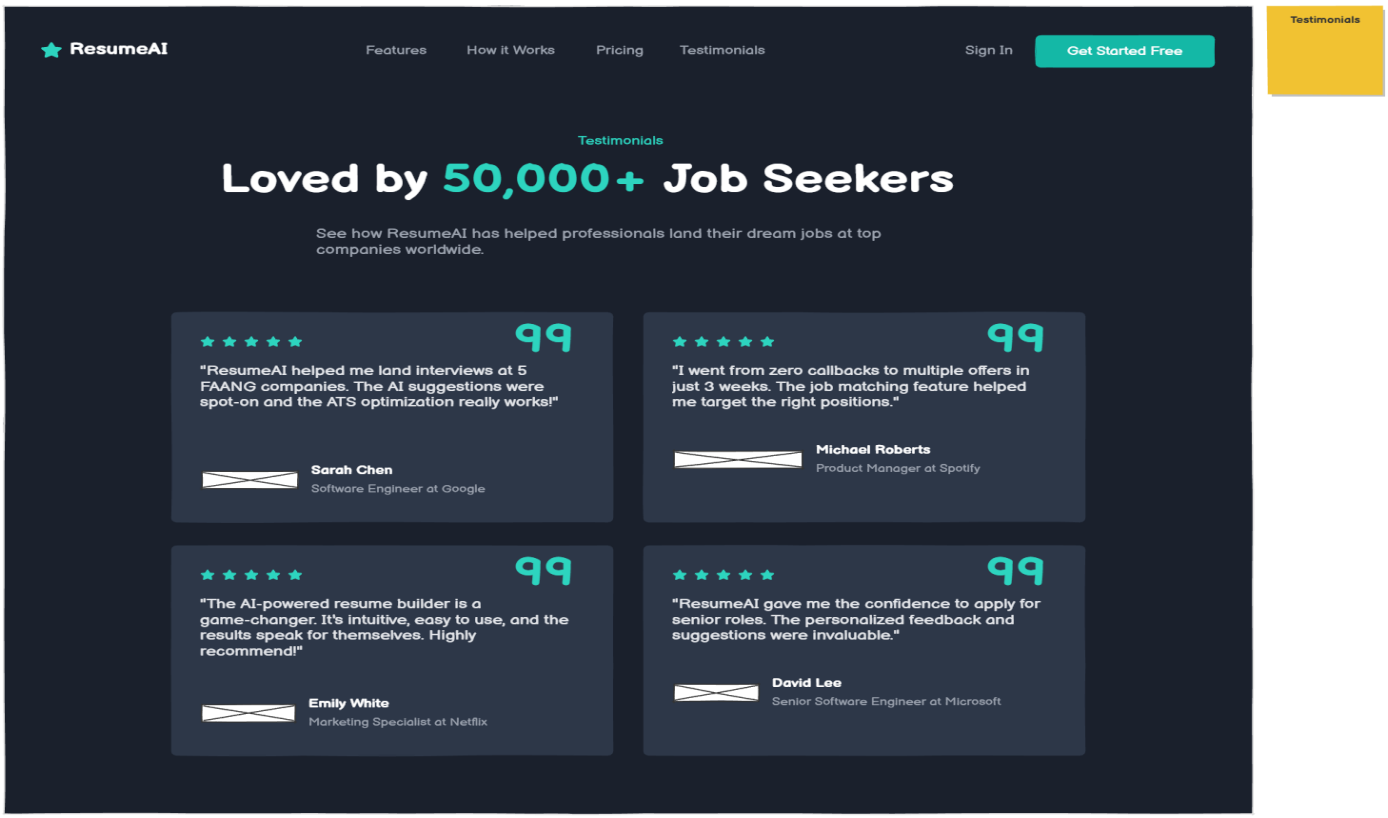

Figure 22 Wireframe of Testimonials Page

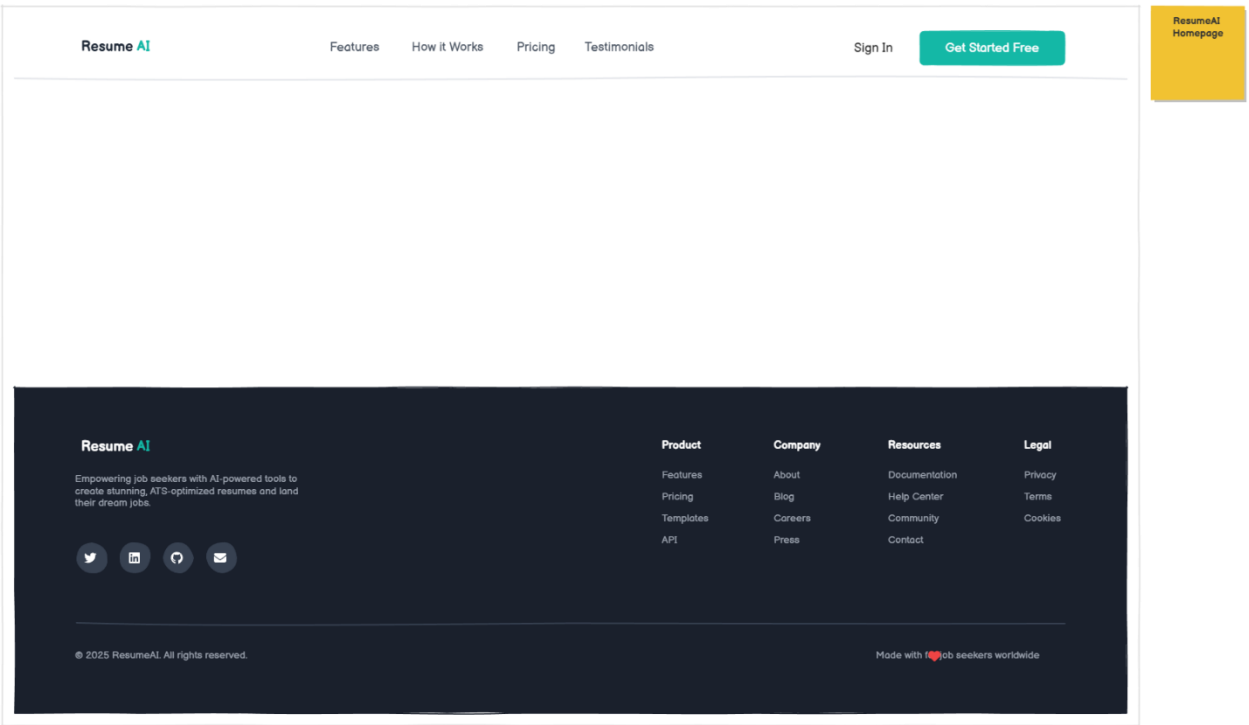

Figure 23Wireframe of Footer page

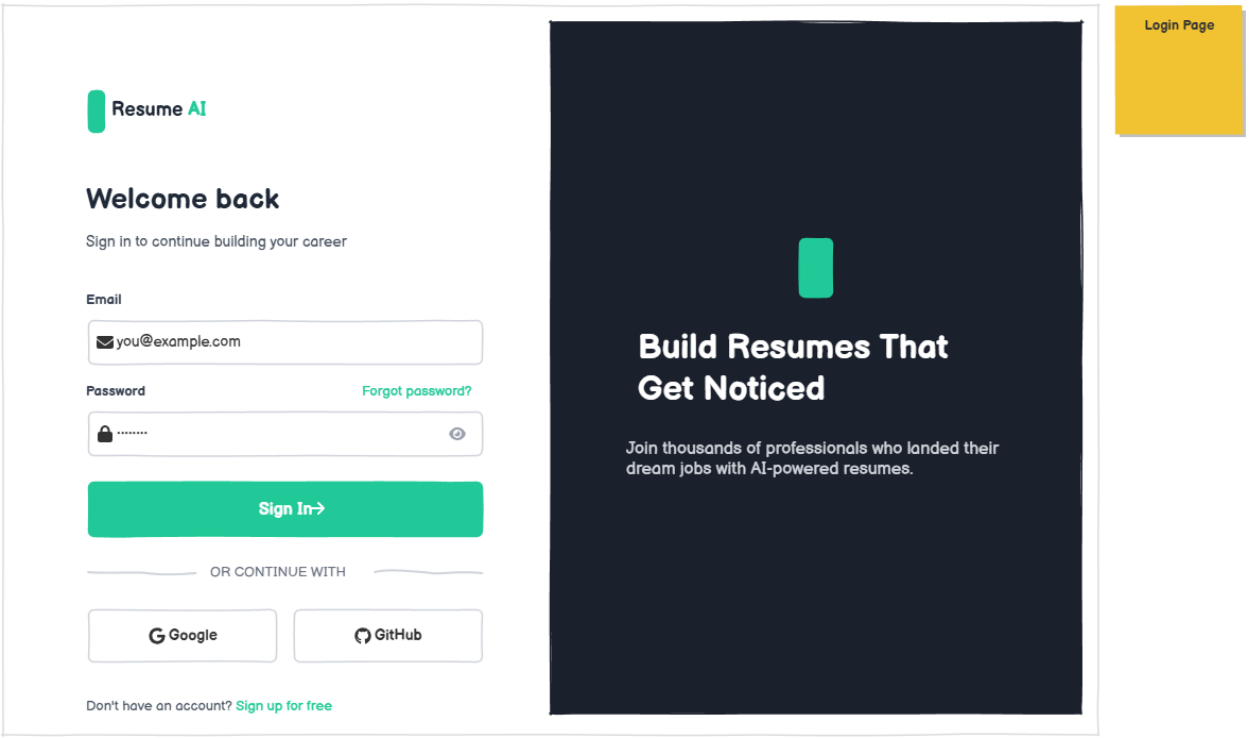

Figure 24 Wireframe of Sign in Page

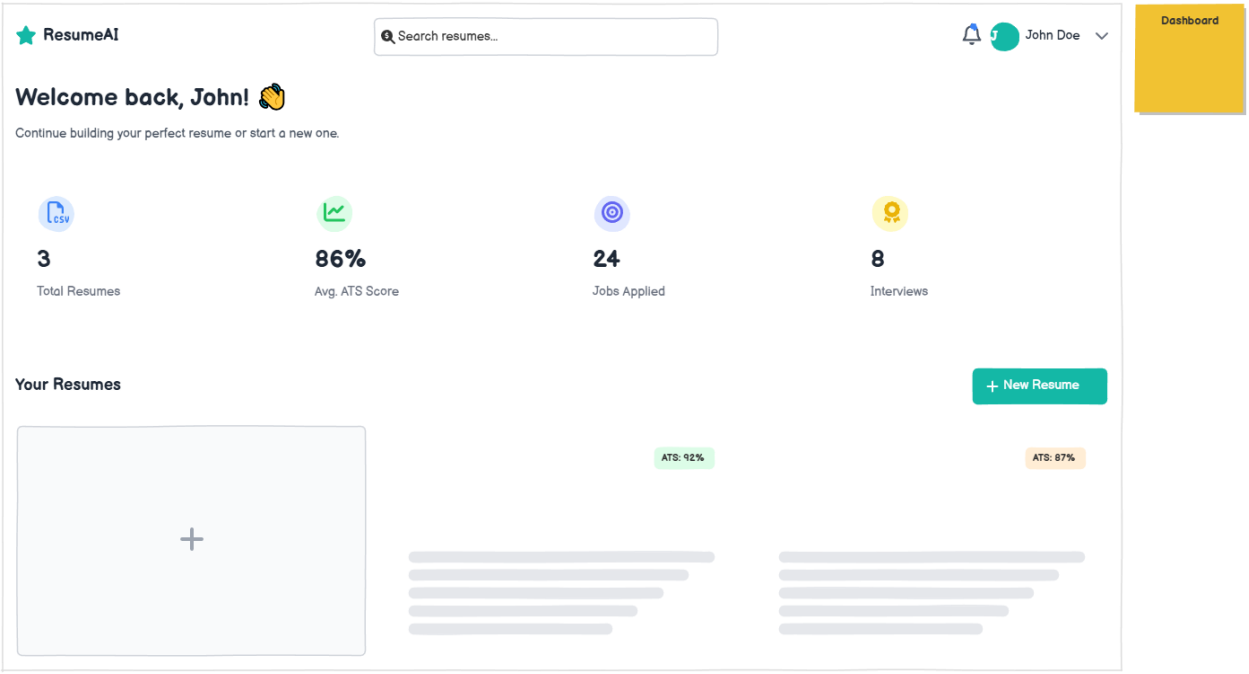

Figure 25 Wireframe of Dashboard

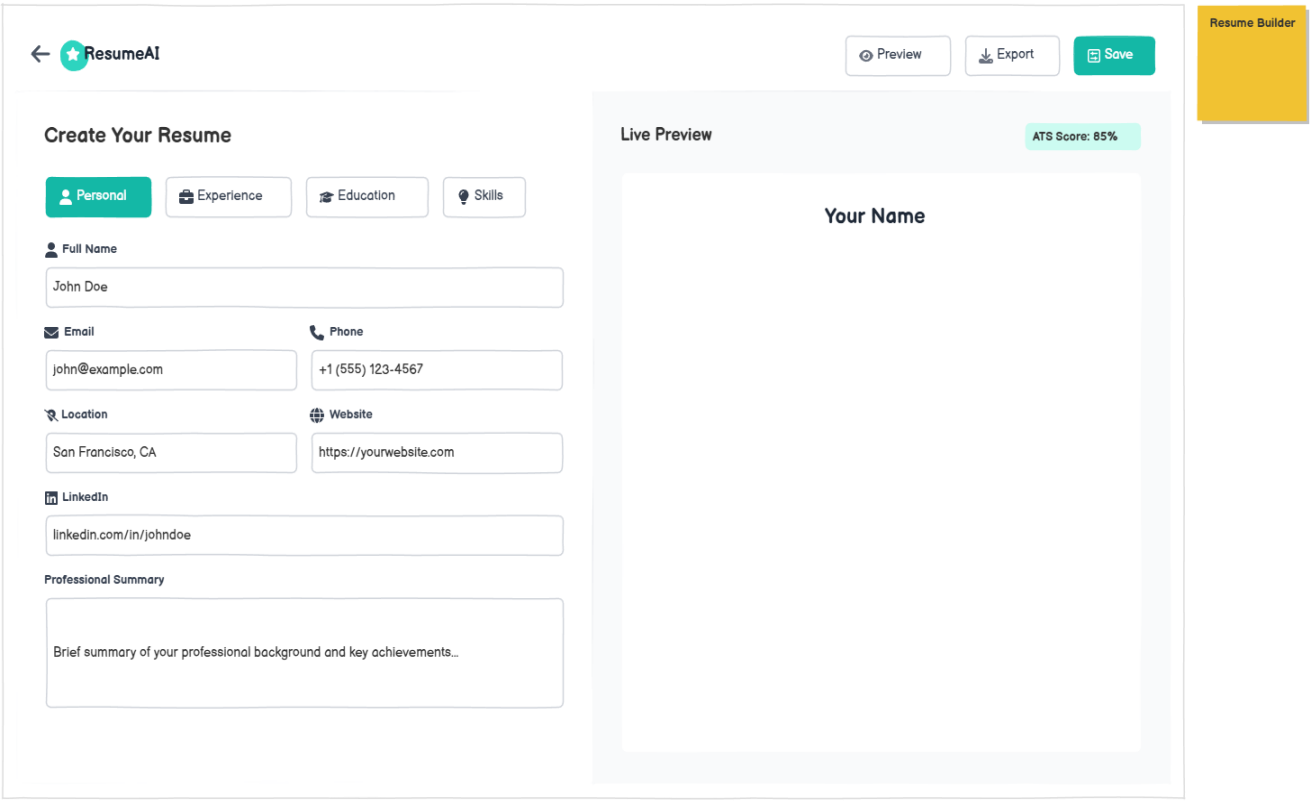

Figure 26 Wireframe of Resume Builder

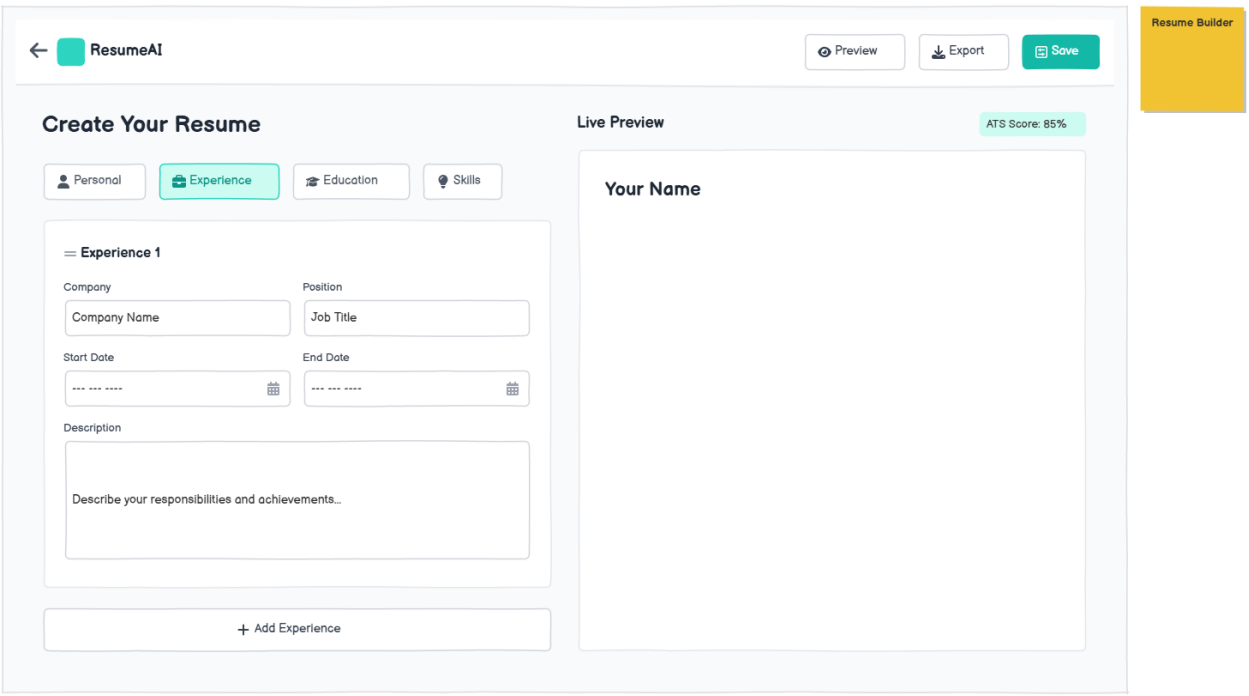

Figure 27 Wireframe of Resume Builder2

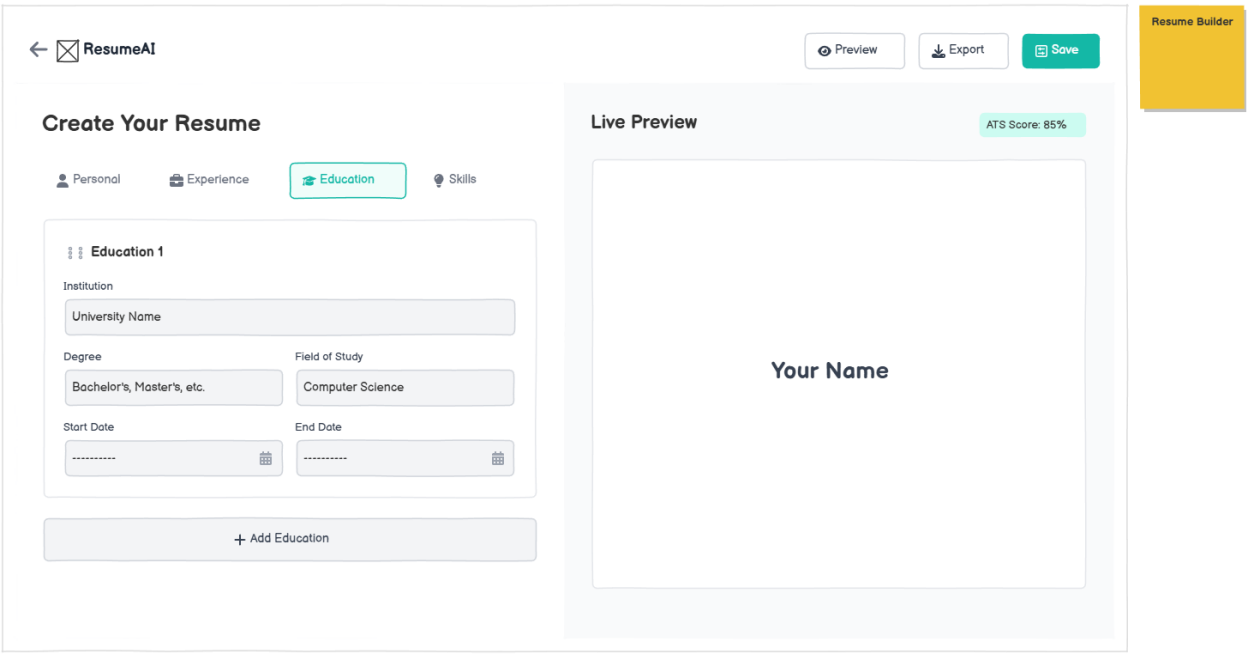

Figure 28 Wireframe of Resume Builder3

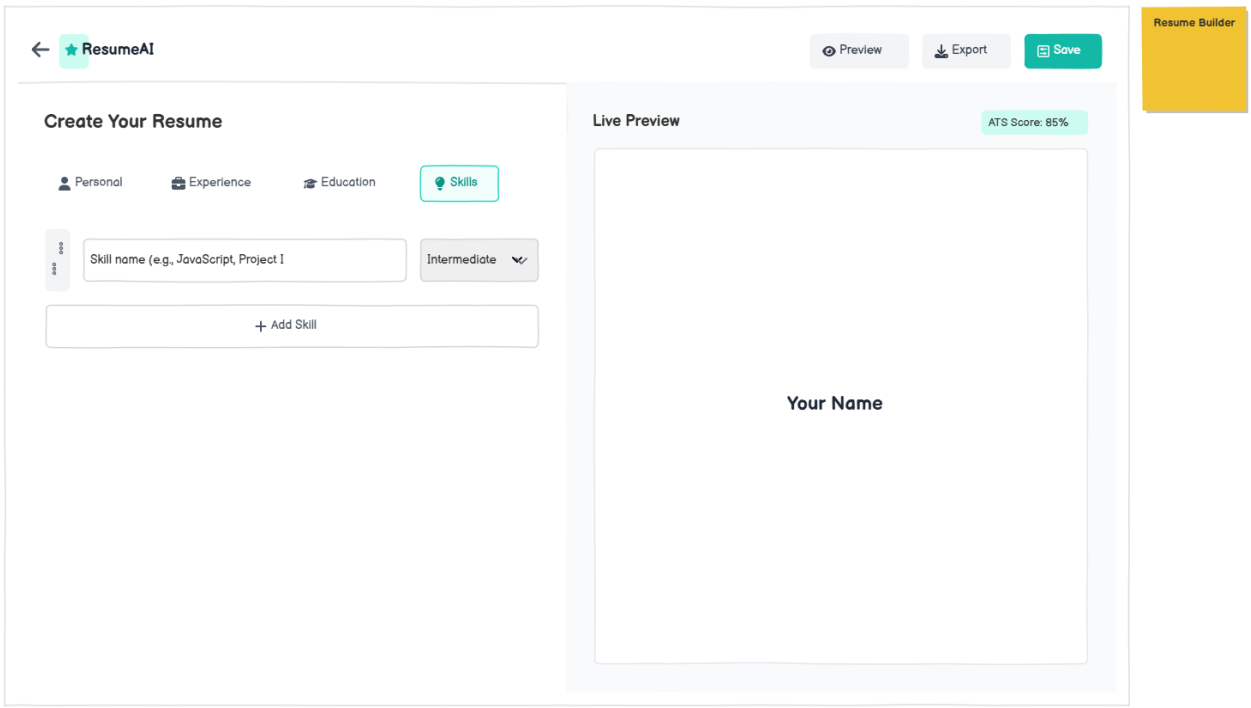

Figure 29 Wireframe of Resume Builder4

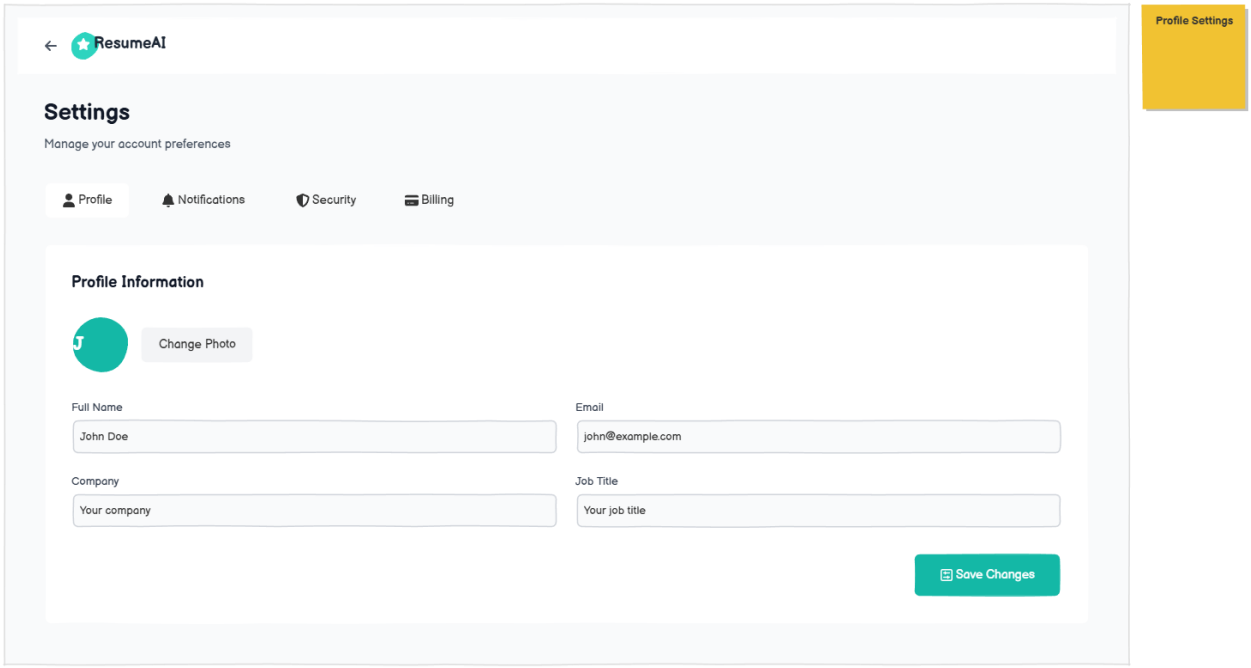

Figure 30 Wireframe of Profile Setting Page

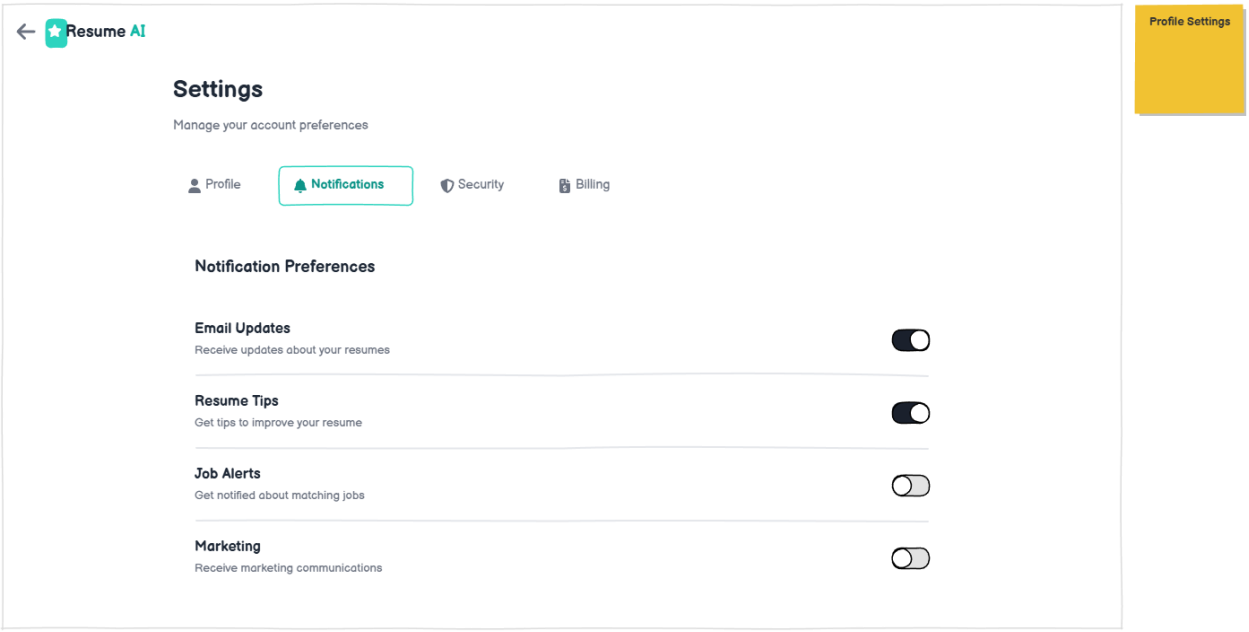

Figure 31 Wireframe of Profile Setting Page2

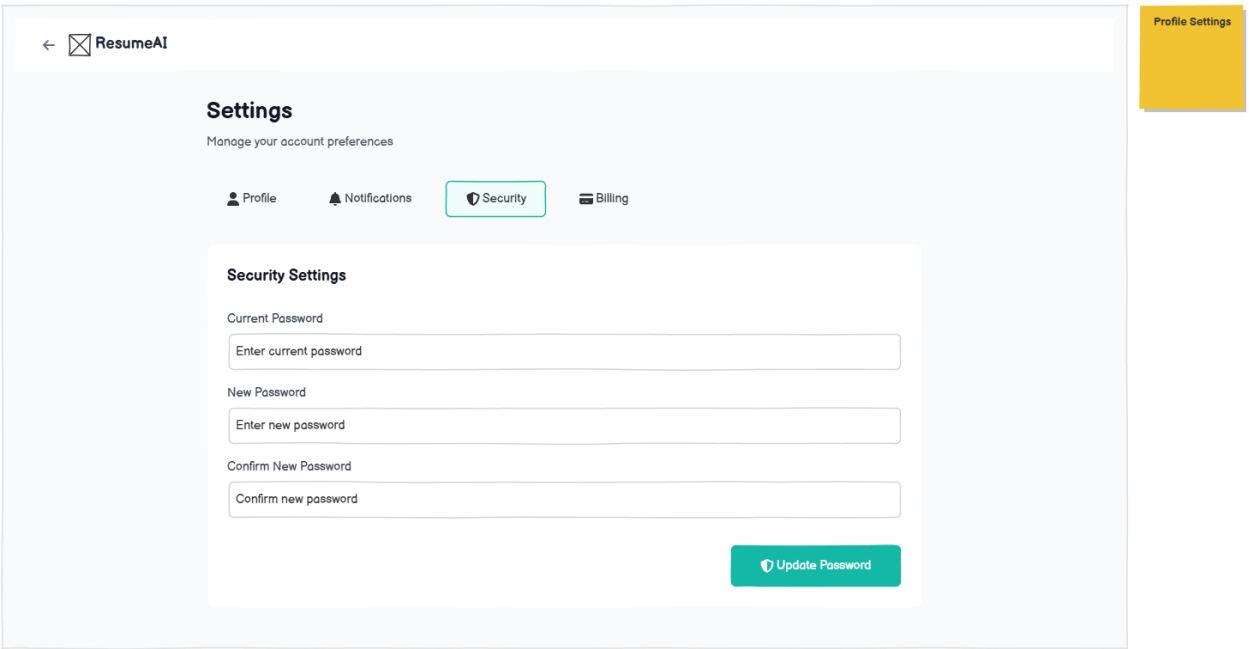

Figure 32 Wireframe of Profile Setting Page3

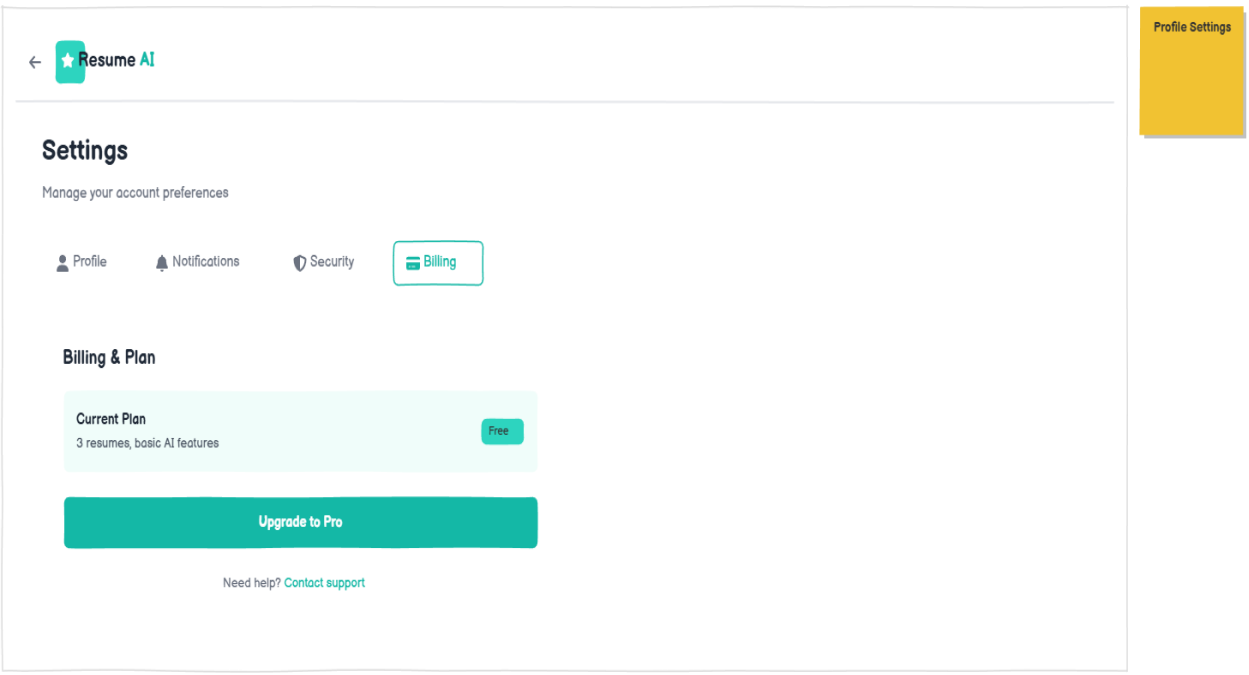

Figure 33 Wireframe of Profile Setting Page4

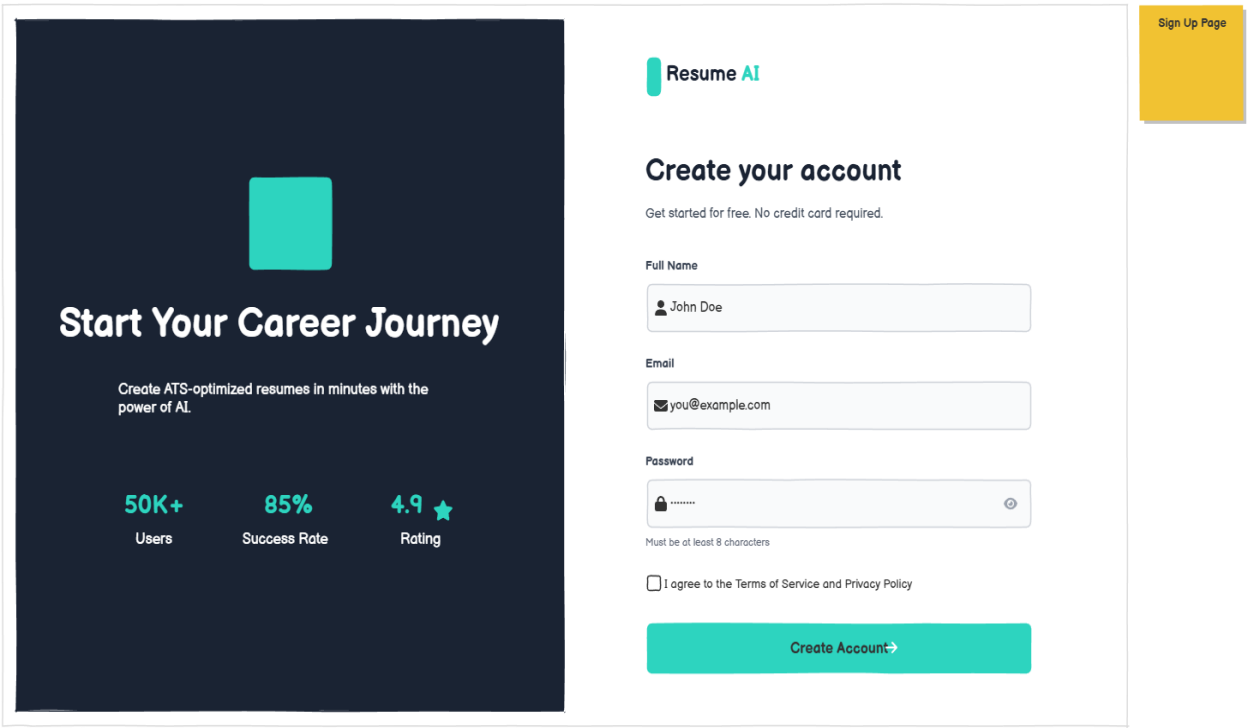

Figure 34 Wireframe of Sign Up Page

References

Garrett, J. J. (2011). *The elements of user experience.* New Riders.

Hamilton, J. (2021). *Software Testing Approaches and Techniques.* Tech Publications.

Jekyll, T. (2018). *Functional decomposition diagram in software systems.* Software Engineering Journal.

Larman, C. (2005). *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development.* Prentice Hall.

Myers, G. J. (2011). *The art of software testing.* John Wiley and Sons.

Pressman, R. S. (2020). *Software engineering: A practitioner's approach.* McGraw Hill Education.

Schwaber, K. a. (2020). *The Scrum Guide.* Scrum.org.

Sommerville, I. (2016). *Software engineering.* Pearson Education.

UX, E. (2021). *What is a wireframe?* Experience UX.